1. Define Cloud Computing and its core characteristics.

Answer: Cloud computing is the on-demand delivery of IT resources (like servers, storage, databases, and networking) over the internet with pay-as-you-go pricing. According to NIST, it has five essential characteristics:

- **On-demand self-service:** Users can provision resources automatically without human intervention from the provider.
- **Broad network access:** Capabilities are available over the network and accessed through standard mechanisms (mobile phones, tablets, laptops).
- **Resource pooling:** The provider's resources are pooled to serve multiple consumers using a multi-tenant model.
- **Rapid elasticity:** Resources can be elastically provisioned and released to scale rapidly outward and inward with demand.
- **Measured service:** Cloud systems automatically control and optimize resource use by leveraging a metering capability (e.g., storage, processing, bandwidth).

2. Compare the three main Cloud Service Models (IaaS, PaaS, SaaS).

Answer: These models define the level of control you have over your infrastructure.

Model	**Full Name**	**Description**	**User Manages**
IaaS	Infrastructure as a Service	Provides fundamental computing resources.	OS, Middleware, Data, Apps
PaaS	Platform as a Service	Provides a framework for developers to build apps.	Applications and Data
SaaS	Software as a Service	Delivers software applications over the internet.	Only basic settings/usage

3. Explain the difference between Public, Private, and Hybrid Clouds.

Answer: * **Public Cloud:** Owned and operated by third-party cloud service providers (like AWS or Azure). Resources are shared with other organizations.

- **Private Cloud:** Used exclusively by a single organization. It can be physically located at the company's on-site datacenter or hosted by a third-party provider.
- **Hybrid Cloud:** A computing environment that combines a public and a private cloud, allowing data and applications to be shared between them. This provides greater flexibility and optimization of existing infrastructure.